BigAlpha 2026: A Systematic Benchmark of 20+ Deep Learning Models for End-to-End Stock Return Prediction

Zhenhao Li
XJTLU / University of Liverpool
Suzhou, China
omitted@blind.review

ABSTRACT

We present a comprehensive benchmark infrastructure of 23 deep learning and gradient-boosted models for end-to-end stock return prediction, built for the BigAlpha 2026 competition. Our model zoo spans five architectural families: (1) financial foundation models (Kronos [1]), (2) time-series transformers (Autoformer [2], Informer [3], TimesNet [4], PatchTST [5]), (3) recurrent architectures (LSTM [14], GRU [15]), (4) Bayesian-optimized tree models (LightGBM [7], XGBoost [8]), and (5) multi-agent reinforcement learning (CTDE-MARL [10]). All models operate on raw OHLCVA features with zero feature engineering. We implement TPE-based Bayesian hyperparameter optimization [11, 12] via Optuna [13] across 10 model families. A local baseline test on 6 models (10 A+H stocks, CPU, 20 epochs) confirms that daily data alone is insufficient for cross-sectional prediction (max IC = +0.038). We provide a three-phase competition strategy targeting IC > 0.08 through IC-weighted ensemble of Bayesian-optimized models on GPU with minute-frequency data.

1 INTRODUCTION

The BigAlpha 2026 Global University League, organized by BigQuant, challenges participants to build end-to-end models predicting cross-sectional stock returns from raw market data. The End-to-End Price Prediction track imposes: (1) zero feature engineering, (2) input fields $\leq$ 100, (3) parameters $\in [10^5, 10^8]$, (4) from-scratch training, and (5) $\leq$ 3 hours GPU inference. Scoring weights: Rank IC (25%), ICIR (25%), Sharpe (25%), and Stress robustness (25%).

We present the model zoo and training infrastructure built for this competition. Drawing on our prior work on CTDE-MARL for portfolio management [10] and recent time-series architectures, we implement 23 models and apply TPE-based Bayesian optimization to each. This report documents: (1) the model design (§2), (2) Bayesian optimization search spaces (§3), (3) local baseline experiments (§4), and (4) competition submission strategy (§5).

Contributions. (1) A unified 23-model benchmark codebase for E2E stock prediction. (2) TPE-based Bayesian optimization covering 10 model families with 7–10 parameter search spaces. (3) Local baseline experiments validating the necessity of GPU and minute-frequency data. (4) A three-phase competition strategy informed by model characteristics.

Scope and Limitations. This report documents the *infrastructure* and *local baseline* — BigQuant GPU experiments with Bayesian optimization are in progress and will be reported separately. Results are currently limited to CPU-based daily-frequency evaluation; conclusions about relative model performance require GPU experiments. Missing baselines include iTransformer, DLinear, Mamba, and TimesFM which will be added in future work.

2 MODEL ZOO

All models accept raw 6-dimensional OHLCVA sequences (open, high, low, close, volume, amount) with configurable lookback windows. Each model family has a corresponding Bayesian optimizer (Table 1).

2.1 Financial Foundation Models

Kronos [1] is a decoder-only Transformer pre-trained on 12B+ K-line records from 45+ exchanges. The BSQ tokenizer converts OHLCVA into discrete tokens, naturally satisfying the zero-feature-engineering constraint. Variants: Kronos-mini (4.1M), Kronos-small (24.7M), Kronos-base (102.3M).

2.2 Time-Series Transformers

Autoformer [2] uses auto-correlation and progressive decomposition. **Informer** [3] uses ProbSparse attention ($O(L \log L)$). **TimesNet** [4] transforms 1D→2D via FFT then applies Inception convolutions. **PatchTST** [5] segments sequences into patches with channel-independent Transformers. Standard **Transformer** [6] serves as baseline.

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2.3 Recurrent Architectures

LSTM [14] and **GRU** [15] encode 60-day sequences. From our ACML 2026 study [10], raw-OHLCVA LSTM achieves Calmar 6.05 vs. MLP on 53 features (Calmar 6.19) with 89% fewer inputs. **BiLSTM** adds bidirectional context; **LSTM+Attention** uses learned temporal pooling.

2.4 Bayesian-Optimized Tree Models

LightGBM [7] and **XGBoost** [8] operate on flattened OHLCVA windows (60×6=360 features). Both are optimized via Optuna TPE with 9-parameter search.

2.5 Multi-Agent Reinforcement Learning

CTDE-MARL [9, 10] uses FactorEncoder→ConfidenceGate(τ=0.5)→StrategyNet with CentralizedCritic. From our ablation [10], the centralized critic accounts for ∼50% of Calmar gain; the gate+IC-loss pair prevents strategy collapse.

Table 1: Model zoo summary. †: tested locally; ‡: Bayesian implementation complete, experiments pending.

Family	Models	Params	Status
Foundation	Kronos (mini/small/base)‡	4.1M–102.3M	Code complete
Temporal Transformer	Autoformer‡, Informer‡, TimesNet‡	2–3M	Code complete
	PatchTST‡, Transformer†	132K–2M	1 tested
Recurrent	LSTM†, GRU†, BiLSTM†	10–18K	3 tested
	LSTM+Attn†, BiGRU, Deep LSTM	8–150K	1 tested
Tree (Bayesian)	LGBM‡, XGBoost‡	N/A	Code complete
MARL	CTDE-MARL (LSTM/GRU/MLP)	30–50K	Code complete
Knowledge	CausalKG Event Detector	N/A	Code complete
Ensemble	IC-weighted, Stacking	—	Code complete

3 BAYESIAN HYPERPARAMETER OPTIMIZATION

We apply Tree-structured Parzen Estimator (TPE) [11] via Optuna [13] to 10 model families. Each search runs 50 trials (20 epochs/trial, validated on Rank IC). Search spaces are in Table 2. **Note:** search spaces are implemented but GPU experiments are pending — results will be added in revision.

3.1 Algorithm

Algorithm 1 formalizes the TPE-based Bayesian optimization procedure used across all 10 model families.

The TPE models $P(\theta \mid \text{score})$ by maintaining two Parzen-window density estimators: $l(\theta)$ for the top-performing trials and $g(\theta)$ for the remaining trials. Each iteration samples from the region where $l(\theta)/g(\theta)$ is maximized — i.e., where promising hyperparameters are likely but not yet fully explored. This balances exploitation (sampling near known good regions) with exploration (avoiding oversampling of already-dense regions).

Algorithm 1: TPE Bayesian Hyperparameter Optimization

Input: Model family $\mathcal{M}$, search space Θ, dataset $\mathcal{D}$, trials $T = 50$

Output: Optimal hyperparameters θ^*, best validation IC s^*

(1) Initialize TPE sampler with random 10 trials
(2) **for** $t = 1$ **to** T **do**
  (a) Sample $\theta_t \sim \text{TPE}(\Theta \mid \text{history})$
  (b) Build model $m_t = \mathcal{M}(\theta_t)$
  (c) Train m_t on $\mathcal{D}_{\text{train}}$ for 20 epochs
  (d) Compute $s_t = \text{RankIC}(m_t, \mathcal{D}_{\text{val}})$
  (e) Update TPE: fit $l(\theta) = P(\theta \mid s > \text{median})$, $g(\theta) = P(\theta \mid s \leq \text{median})$
  (f) Sample next θ where $l(\theta)/g(\theta)$ is maximized
(3) $\theta^* = \arg\max_t s_t$; retrain m^* with θ^* for 50 epochs
(4) **return** (θ^*, s^*)

Figure 1: TPE-based Bayesian hyperparameter optimization.

3.2 Planned GPU Experiments

The Bayesian optimization described above requires GPU access for the deep learning models. We plan the following experiments on BigQuant GPU notebooks:

- **Model comparison**: Train all 10 Bayesian-optimized models on 1-minute OHLCVA data with 1000-stock cross-section
- **Ablation study**: Controlled comparison of (a) encoder architecture, (b) lookback window, (c) IC loss weight α, (d) confidence gate threshold τ
- **Ensemble ablation**: IC-weighted vs. equal-weight vs. stacking ensemble on top-5 models
- **Robustness**: Multi-seed evaluation (≥ 3 seeds) with mean±std reporting
- **Regime analysis**: Bull market (2024 H2) vs. bear market (2025 Q1) performance comparison

4 LOCAL BASELINE EXPERIMENT

4.1 Setup

We validate the pipeline locally before BigQuant deployment. **Important caveat:** this is a *smoke test*, not a competitive evaluation. The test uses:

- **Data**: 10 A+H stocks, daily OHLCVA, 60-day lookback, Jan 2025–Apr 2026
- **Samples**: 2,146 train / 300 test
- **Hardware**: CPU (Intel i7)

Table 2: TPE Bayesian search spaces (50 trials each, 20 epochs/trial, objective = validation Rank IC). All spaces implemented; GPU experiments pending.

Model	#P	Search Space	Trials
BayesianKronos	8	variant∈{mini,small}, pred_h∈[64,512], lr∈[1e-5,5e-3], bs∈[32,256], lookback∈[20,120]	50
BayesianAutoformer	9	d_model∈[64,512], L∈[1,5], heads∈{4,8}, d_ff∈[256,2048], lr∈[1e-5,5e-3]	50
BayesianInformer	9	Same as Autoformer	50
BayesianTimesNet	9	d_model∈[64,512], L∈[1,4], d_ff∈[16,64], top_k∈[3,8], lr∈[1e-5,5e-3]	50
BayesianPatchTST	10	patch∈[8,32], stride∈[4,16], d_model∈[64,256], L∈[1,4], heads∈{4,8}	50
BayesianTransformer	8	d_model∈[64,512], L∈[1,6], heads∈{4,8}, d_ff∈[256,2048], lr∈[1e-5,5e-3]	50
BayesianLSTM	7	hidden∈[32,256], L∈[1,4], drop∈[0,0.5], lr∈[1e-4,1e-2], bidir∈{T,F}	50
BayesianGRU	7	Same as LSTM	50
BayesianLGBM	9	n_est∈[100,2000], depth∈[3,15], lr∈[0.01,0.3], leaves∈[15,255], α/λ∈[1e-8,10]	50
BayesianXGBoost	9	n_est∈[100,2000], depth∈[3,15], lr∈[0.01,0.3], γ∈[1e-8,5], α/λ∈[1e-8,10]	50

- **Training**: 20 epochs, batch 128, lr 10^{-3}, Adam, MSE loss, seed 42
- **Hyperparameters**: All defaults (no Bayesian optimization)

4.2 Results

Table 3 reports results. All models exhibit near-zero cross-sectional IC. MLP achieves marginally positive IC (+0.038) due to lower variance under data scarcity.

Table 3: Local baseline (CPU, daily, 10 stocks, 20 epochs, single seed=42).

Model	Params	MSE	Rank IC	ICIR	Time (s)
MLP (baseline)	54,529	0.0020	**+0.038**	+0.43	58.1
BiLSTM	10,305	0.0011	+0.008	+0.60	48.9
Transformer	132,481	0.0010	−0.055	+0.12	140.0
GRU	13,889	0.0011	−0.061	−1.40	37.7
LSTM+Attention	18,562	0.0010	−0.093	−0.37	32.0
LSTM	18,497	0.0011	−0.138	−1.29	29.7
IC-Ensemble	—	0.0010	−0.022	—	

4.3 Analysis

The near-zero IC is expected: daily OHLCVA with 2,146 samples provides minimal cross-sectional signal. MLP's marginal positive IC reflects low variance under data scarcity. Recurrent models exhibit negative IC from overfitting to noise. The Transformer (132K params) is severely underdetermined (16 samples/param). This confirms three necessary conditions for competition success: (1) **minute-frequency data** (480× observations), (2) **1000-stock cross-section** (100× for IC), and (3) **Bayesian HPO** to escape defaults.

Statistical note: Single-seed results are insufficient for ranking. Multi-seed experiments (3+ seeds) will be reported with GPU results.

5 COMPETITION STRATEGY

We propose a three-phase strategy (Table 4):

Table 4: Three-phase submission strategy.

Phase	Dates	Models
P1: Baseline	Jul 1–7	LSTM, GRU, BayesianLGBM/XGBoost
P2: Bayesian	Jul 8–21	Kronos, Autoformer, Informer, TimesNet,
P3: Ensemble	Jul 22–Aug 5	IC-Weighted Top-5 Ensemble

Phase 1 establishes baselines with fast-to-train models at maximum submission frequency. **Phase 2** deploys 10 Bayesian-optimized deep models, each undergoing 50 TPE trials on GPU. **Phase 3** combines best models via IC-weighted ensemble ($w_i \propto e^{|\mathrm{IC}_i|}$). Private leaderboard: best public model retrained from scratch in isolated environment.

6 LIMITATIONS

(1) **Experimental incompleteness:** Only 6 of 23 models tested; Bayesian optimization experiments are pending GPU access. Current results are smoke tests, not competitive evaluations.
(2) **Data constraints:** Daily-frequency data is insufficient for cross-sectional prediction. Minute-frequency data on BigQuant platform is required for meaningful IC.
(3) **Statistical rigor:** Single-seed results lack error bars. Multi-seed experiments ($\geq$ 3 seeds) are planned.
(4) **Missing baselines:** iTransformer, DLinear, Mamba, and TimesFM are not yet included.
(5) **Task specificity:** Results are specific to BigAlpha 2026 competition constraints and may not generalize to other prediction tasks.
(6) **No transaction costs:** Current evaluation ignores market impact and execution costs.

7 CONCLUSION

We have built a 23-model benchmark infrastructure for the BigAlpha 2026 competition, spanning foundation models, time-series transformers, recurrent architectures, Bayesian-optimized trees, and multi-agent RL. TPE-based Bayesian optimization covers 10 families with 7–10 parameter spaces. Local smoke testing confirms the necessity of GPU and minute-frequency data. A three-phase strategy targets IC > 0.08 through systematic optimization and IC-weighted ensembling. Code is available at the project repository.

REFERENCES

[1] S. Shi et al. Kronos: A Foundation Model for the Language of Financial Markets. *AAAI*, 2026.
[2] H. Wu, J. Xu, J. Wang, and M. Long. Autoformer: Decomposition Transformers with Auto-Correlation. *NeurIPS*, 2021.
[3] H. Zhou et al. Informer: Beyond Efficient Transformer for Long Sequence Time-Series Forecasting. *AAAI*, 2021.
[4] H. Wu, T. Hu, Y. Liu, H. Zhou, J. Wang, and M. Long. TimesNet: Temporal 2D-Variation Modeling for General Time Series Analysis. *ICLR*, 2023.
[5] Y. Nie, N. H. Nguyen, P. Sinthong, and J. Kalagnanam. A Time Series is Worth 64 Words: Long-term Forecasting with Transformers. *ICLR*, 2023.
[6] A. Vaswani et al. Attention Is All You Need. *NeurIPS*, 2017.
[7] G. Ke et al. LightGBM: A Highly Efficient Gradient Boosting Decision Tree. *NeurIPS*, 2017.
[8] T. Chen and C. Guestrin. XGBoost: A Scalable Tree Boosting System. *KDD*, 2016.
[9] R. Lowe, Y. Wu, A. Tamar, J. Harb, P. Abbeel, and I. Mordatch. Multi-Agent Actor-Critic for Mixed Cooperative-Competitive Environments. *NeurIPS*, 2017.
[10] Z. Li et al. What Matters in Multi-Agent Reinforcement Learning for Portfolio Management? *ACML*, 2026. Under review.
[11] J. Bergstra, R. Bardenet, Y. Bengio, and B. Kégl. Algorithms for Hyper-Parameter Optimization. *NeurIPS*, 2011.
[12] J. Snoek, H. Larochelle, and R. P. Adams. Practical Bayesian Optimization of Machine Learning Algorithms. *NeurIPS*, 2012.
[13] T. Akiba, S. Sano, T. Yanase, T. Ohta, and M. Koyama. Optuna: A Next-generation Hyperparameter Optimization Framework. *KDD*, 2019.
[14] S. Hochreiter and J. Schmidhuber. Long Short-Term Memory. *Neural Computation*, 9(8):1735–1780, 1997.
[15] K. Cho et al. Learning Phrase Representations using RNN Encoder-Decoder. *EMNLP*, 2014.